A Lever, Not a Ruler: Steering an Output Modality You Cannot Address

Abstract

You can address the generator in one modality and you need the output in another. You write text; a decoder you did not train turns it into a program that runs, a query that returns rows, a picture. The pipeline then does what is natural: it embeds the text and makes its decisions in that space — drop the near-duplicates, keep the most different items, retrieve, screen for contamination — because the text is the only thing it can touch.

This paper measures what that channel can and cannot do, on decoders that are deterministic and total, so artifact distance is a property of the text alone and repeats exactly. The answer is a clean split, and it is the same channel in both halves.

The channel carries real information about the artifact. Permuting the artifacts across items destroys the association in every domain we measure (near-minus-far −0.025 against +0.386 unpermuted; pooled −0.003 against +0.364). Text and artifact are genuinely coupled.

Its geometry is not a measure of that artifact. The standard way to qualify a proxy — decode a sample, correlate, observe that agreement is strongest among similar items — is confounded. Running the identical analysis between two *encoders*, with no decoder anywhere, reproduces the effect: near-minus-far +0.742 to +0.781 against the decoder's +0.386 on generated Python programs. On three decoders, three artifact metrics and five encoders, **no decoder profile clears its encoder-to-encoder baseline distribution**. The one apparent exception failed on inspection: bootstrapped over items, its margin is −0.195, 95% CI [−0.337, +0.013]. Downstream, the decisions built on that geometry come apart — deduplication ranks duplicate pairs at AUC 0.779 but **45.4% of the pairs it would reject are behaviourally far apart**, and greedy max-min loses to random on one decoder while beating it on another, so its sign cannot be assumed.

The same channel works as a lever. Stating the target in the text — `f([1, 2, 3])` *must return 7* — and checking the result by execution: compliance **62.5%** [54.2, 70.1], against **6.9%** [2.8, 11.1] for a decoy arm that states a different target in identical form, and 18.1% for no instruction at all. The decoy sits *below* blind prompting, so the channel transmits which target rather than merely transmitting specificity. Compliance is a function of where the target sits in the generator's own artifact distribution: **100% in the head, 30.6% in the tail.**

So measure with the decoder and steer with the channel. For deterministic decoders the artifact-space oracle is cheaper than the embeddings it replaces — 29.1 ms to decode a program in its own OS process against 107.9 ms to embed one, 0.98 ms against 685 ms on a second decoder — and it buys 1.54× [1.29, 1.85] the coverage of text-space selection at $k = 10$, needing a median of 30 items where max-min needs 100 (IQR [100, 150]).

We ship this as `decodergap`, which reports the decoder-free control beside every correlation and scores each decision in the artifact's own space. The last section treats the case this advice does not reach: rendered images and audio, where decoding is expensive, text-embedding similarity explains 14.5% of the variance in rendered-image similarity, and a third of the conditioning is lost at the seam where one model's language becomes another model's input.

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1. Introduction

A generated corpus is rarely the thing anyone wants. It is a step towards the thing: the program that will run, the query that will return rows, the picture that will be rendered. Between the text and the artifact sits a **decoder** — an interpreter, a database engine, a diffusion model — and the generator is addressable only on the near side of it. You write text. You get something else.

Pipelines respond by working in the modality they can touch. They embed the text and make their decisions with those distances: deduplicate at a threshold, keep the k most mutually distant items, retrieve neighbours, screen an evaluation set against a training set for leakage. Each of those is a claim that the geometry of the control modality stands in for the geometry of the output modality — a claim which is almost never tested, because testing it appears to require decoding everything, which appears to be expensive.

This paper tests it, and the result is a split rather than a verdict. **The control modality is a lever, not a ruler.**

As a *ruler* — measure structure in the text, infer structure in the artifact — it fails, and it fails in a way that is invisible to the check people run. As a *lever* — write an instruction and let the generator carry it across — it reaches the artifact, with a compliance rate we can quantify and a boundary we can locate. Both halves are measured on the same corpora, with the same generator and the same encoders, so the contrast is not an artifact of comparing two literatures.

The hinge between them is a number we nearly ignored. Permuting the artifacts across items destroys the text-to-artifact association completely, in every domain. **The information is there.** What fails is reading it off a metric defined on the text. That distinction is the whole paper: a channel can be informative without its geometry being a measurement.

Why deterministic decoders. The companion paper, *Recursive Axis Conditioning for Diverse Synthetic Data Generation*, builds a generation loop on an embedding oracle with no inverse, and its artifact-space claims are bounded by something no care removes: the image endpoint exposes no seed, so no render can be repeated and render noise cannot be separated from what is being measured. Choosing decoders that are deterministic and total removes that bound. A Python program on a fixed battery of inputs, a query against a fixed database, a pattern against a fixed corpus of strings — each repeats exactly, so artifact distance is a property of the text alone.

What we found by being wrong. The first version of this paper reported a mechanism: embeddings as near-field instruments, tracking the artifact among near-duplicates and saying nothing beyond. It did not survive its own control. Four further claims died the same way, every one of them built on the shape of a correlation, and every claim built on running a decision and scoring it in the artifact's space has held. §8 reports that pattern, because it is the paper's own thesis applied to the paper's own methodology, and it is the best argument we have that the confound is the default rather than an exotic failure mode.

The contributions, in the order we would defend them:

- **The lever result** (§4): a target stated in the control modality is hit 62.5% of the time against 6.9% for a matched decoy, with a compliance curve that falls from 100% to 30.6% as the target moves into the generator's tail.
- **The ruler result** (§3): the confound that invalidates correlational proxy validation, on three decoders and five encoders, with the control that detects it and the permutation null that shows

what a real association looks like.

- **The decision verdicts** (§3.5–§3.7), each a direct measurement: aggregate deduplication supported, per-item refused at a 45.4% false-positive rate, diversity selection sign-unstable, and 5.2× behavioural redundancy surviving perfect text-level deduplication.
- **The cost inversion** (§5): for deterministic decoders the artifact-space oracle is cheaper than embedding the corpus, with bootstrapped intervals on what it buys.
- **`decodergap`** (§6), which runs all of the above for a corpus, an embedder and a decode function, and declines to recommend a threshold.
- **The expensive-decoder case** (§7): what remains when decoding cannot be afforded, measured on rendered images and audio.

2. Setup

Generator and embedders. `openai/gpt-5.6-luna` via OpenRouter for generation, judging, attractor mining and axis elicitation. Text embeddings: `nomic-embed-text` (768-d, unit-normalized) served locally by Ollama, so embedding is free and the loop is never rate-limited by its own measuring instrument. Images: `gpt-image-1-mini`, embedded with CLIP ViT-B/32. Audio: Lyria 3 Pro instrumentals, embedded with CLAP and MERT. Every corpus is append-only JSONL with embeddings checkpointed alongside, resumable after a kill; every number carries a provenance tag naming the procedure that produced it.

Domains. *DALL·E instructions for post-modern artworks*, where diversity is the product and the same corpus can be measured in three spaces — words, text-embedding, and the rendered picture. *Psychometric test questions*, multiple-choice items assessing general reasoning, where two items with the same construct are a validity problem rather than an aesthetic one. *Instruction corpora*, for the head-to-head against released artifacts. *Poetry*, a pilot where craft rather than content carries the variation. *Instrumental-music prompts*, the longest chain in the paper: latent axes → text prompt → two-minute instrumental → audio embedding.

Methods compared. Five published approaches, implemented faithfully rather than as strawmen, each getting the same generator, embedder and budget accounting as ours.

arm	what it is
`naive`	plain repeated prompting — the honest floor
`high_temp`	temperature 1.6 — the trivial diversity lever
`self_instruct`	**Self-Instruct / Alpaca**: few-shot exemplars sampled from the pool, plus ROUGE-L ≤ 0.7 rejection against it
`evol_instruct`	**Evol-Instruct (WizardLM)**: sample an existing item, apply a random evolution operator (deepen, concretize, add constraint, harder reasoning, mutate form)
`persona`	**Persona-Hub / AttrPrompt**: a flat catalogue of personas × attributes, sampled uniformly
`rac`	**Recursive Axis Conditioning** (ours)
`rac+vision`	RAC, plus steering on the *rendered image* (§7.6.1)

What we measure, and what each measure misses. We report several because they disagree, and a corpus that one of them calls healthy another calls collapsed.

- **Exact-duplicate rate** — byte-identical repetition. Unambiguous, cheap, and the first thing to compute; blind to paraphrase.
- **Distinct-*n*, 4-gram self-repetition, *n*-gram Vendi** — surface statistics over token sequences. They catch templating that the duplicate rate misses and read meaning not at all. **These are independent of our objective**, since the method sees only embeddings and has no access to token counts or string identity.
- **Mean-centered embedding Vendi** — the exponential of the von Neumann entropy of the corpus Gram matrix, an effective number of distinct items. It summarizes the whole spectrum, which makes it insensitive to local structure. Centering matters as much as the statistic: the shared mean direction of text embeddings compresses the uncentered score by roughly 2.7×, so an uncentered Vendi reports the cone as much as the content.

- **Median nearest-neighbour distance** — how much room the typical item has. It goes to exactly zero when the median item has a perfect twin.
- **Worst-case nearest-neighbour distance (the packing radius)** — the only measure here under which a single collision is a defect regardless of everything else.
- **Coverage, density, precision and recall against a reference** — computed with reference-side k-NN radii so nothing is tunable per corpus. These are the only measures that know what the space is supposed to look like, and the only ones that let corpora of different scales be compared. Covered fraction at a *fixed* radius does not survive that comparison: it correlates −0.991 with within-corpus spacing, so it ranks corpora by how tightly they cluster rather than by how much they reach.
- **A judged threshold**, where an application defines the failure (the companion paper's §7.4).
- **Measures on the rendered artifact**, and **measures in a channel the loop never optimizes** (§4.1), which are the ones that cannot be circular. **Scale and cost.** The text corpora comprise 43,171 real generations for roughly \$8.50 of OpenRouter spend, plus 1,796 rendered images across sixteen image arms and up to three quality tiers, a further 285 renders for the two render-side probes of §7.6.3 and §7.7, 100 rendered Lyria instrumentals, and 236 adjudicated exam-item pairs. Both `naive` arms reach n = 10,000; the reimplemented baselines reach 2,500 each. Every comparison is reported at a matched n that all compared arms actually reached. Total API spend across the project is roughly \$40.

3. A proxy cannot be validated by correlating it with the artifact

Everything in this paper so far is a statement about diversity. The measurement underneath it is not. A pipeline that embeds text and acts on the distances is making a bet that has nothing to do with diversity in particular: that proximity in the embedding stands in for proximity in the artifact. Deduplication makes it, contamination screening makes it, nearest-neighbour retrieval makes it, coreset selection and active learning make it.

The natural way to check the bet is to decode a sample, compute a distance in the artifact's space, and correlate. This section shows that this check does not work — not that it is noisy, but that it returns the same answer when the artifact is replaced by something with no connection to it — and then measures the three decisions directly instead, which is what survives.

3.1 Decoders that leave nothing to interpretation

An image is rendered by a model, which is why every artifact-space number in the companion paper is a statement about an unlogged seed distribution: the endpoint exposes no seed, so a render cannot be repeated and render noise cannot be separated from the quantity being measured. That bound is structural.

It is removed by choosing decoders that are *deterministic and total*. A Python program run on a fixed battery of inputs, a SQL query run against a fixed database, a regular expression matched against a fixed corpus of strings: each maps text to an artifact with no sampling anywhere, so artifact distance is a property of the text alone and repeats exactly. These are also among the largest synthetic-data industries there are.

domain	text	artifact	artifact distance	coverage
code	a Python function `f(xs)`	its results on a fixed 140-input battery	fraction of the battery on which two programs disagree	distinct (input, result) cells
SQL	a `SELECT` against a fixed schema	the rows it returns	Jaccard distance between result-row sets	distinct rows retrieved
regex	a pattern	the subset of a 210-string probe corpus it matches	Jaccard distance between matched sets	distinct strings matched
library inputs	a string handed to `ipaddress`	the arcs of the library it executes	Jaccard distance between arc sets	distinct arcs reached

None of these distances is computed with reference to an embedding, so each can adjudicate an embedding rather than agree with it. The first is the corpus used below: 553 generated Python programs, of which 203 are distinct sources, which between them exhibit **39 distinct behaviours**.

3.2 The profile that looks like a finding

Correlating text distance against behavioural distance *within* each decile of text distance, rather than pooling, produces a table that appears to say something sharp:

decile of text distance	width	pairs	corr. with behavioural distance	mean behavioural distance
1 (closest)	0.140	2,051	**+0.451**	0.549
2	0.032	2,050	+0.112	0.827
3	0.022	2,050	+0.010	0.877
4–9	0.014–0.018	12,301	−0.040 … +0.062	0.878–0.909
10 (farthest)	0.104	2,051	**+0.065**	0.916
pooled		20,503	+0.364	0.853

Read on its own this says: the embedding tracks behaviour among near-duplicates and stops carrying information beyond them. The conditional mean of behavioural distance given text distance says the same thing without any correlation at all, rising **+0.474 across the near half of the distance range and +0.036 across the far half**. Every diversity selector in the literature draws its picks from the top decile.

We believed this, and it is wrong.

3.3 The control: two encoders, no decoder

The profile compares one distance matrix to another and reports that they agree more among close pairs. Nothing in that description mentions a decoder. If any two distance matrices over the same points agree preferentially in the near field, the shape is a property of the comparison and the decoder is doing no work at all.

Running the identical profile between two *encoders*, with no decoder anywhere in the computation:

comparison	near-field corr.	far-field corr.	**near − far**
nomic vs behaviour *(the claim)*	+0.451	+0.065	**+0.386**
char TF-IDF vs behaviour	+0.474	−0.002	+0.476
word TF-IDF vs behaviour	+0.513	−0.053	+0.567
nomic vs char TF-IDF *(control)*	+0.742	−0.003	**+0.745**
nomic vs word TF-IDF *(control)*	+0.750	+0.007	**+0.742**
char TF-IDF vs word TF-IDF *(control)*	+0.898	+0.117	**+0.781**
nomic vs *permuted* behaviour *(null)*	−0.002	+0.023	−0.025

The decoder-free controls show the shape about twice as strongly as the decoder does, and this holds on every decoder we have measured. A second decoder — 603 generated SQL queries scored by the cells they return — at first appeared to be the exception, with a decoder profile of +0.231 against cross-family baselines of −0.066 and +0.037. It is not. Bootstrapping over items (300 resamples), its margin over its *largest* baseline is **−0.195, 95% CI [−0.337, +0.013]**, positive in 5.0% of resamples. On three decoders now — Python programs, SQL queries, and strings handed to a library — the decoder profile fails to clear the encoder-to-encoder baseline. The confound explains the whole of every correlation we have. The conditional-mean curve is the same story: the control rises +0.493 over the near half and +0.030 over the far half, against the decoder's +0.474 and +0.036 — the two curves are the same curve. Two representations that have never seen a program run produce a cleaner ”near-field law” than the comparison against what the programs actually do.

A third of the effect is bookkeeping. Equal-count deciles have very unequal widths — the first spans 0.140 of the distance range and the eighth spans 0.014 — and a within-bin correlation is attenuated by the bin's own spread. On equal-width bins the profile is +0.288, +0.056, +0.114, +0.191, +0.061, −0.004, +0.014, +0.030, +0.070, −0.080: still more signal at the low end than the high end, and nothing like the monotone decline the decile table displays.

One control does pass, and it is the one that matters for what remains. Permuting the artifacts across items — breaking the text-to-behaviour correspondence while leaving both marginal distance distributions exactly intact — flattens everything to −0.025 near-minus-far and −0.003 pooled, against +0.364 unpermuted. **There is a real association between what a program looks like and what it does.** What there is no evidence for is the shape we read into it.

3.4 The consequence: correlational proxy validation is confounded

This generalizes past our own mistake, because the check we ran is the check the field runs. A paper that proposes an embedding-based proxy for something expensive — human preference, rendered quality, downstream accuracy, behavioural diversity — validates it by decoding a sample and reporting a correlation, and frequently reports that the correlation is strongest among similar items. That last observation is not evidence. It appears between two arbitrary encoders that have never seen the outcome.

The prescription is cheap and almost nobody follows it:

> **Report a decoder-free control beside any proxy-validation correlation.** Correlate your embedding against a second, unrelated representation of the same corpus. Whatever agreement survives *above* that control is the part your artifact measurement contributed. On the corpus here, that residue is negative: the decoder profile is weaker than the encoder-to-encoder baseline.

And the positive form: **validate a proxy by running the decision, not by correlating the distances.** A correlation is a summary of a geometry; a decision is a thing that either works or does not, is scored in the artifact's space, and has no confound of this kind. The rest of this section does that for the three decisions a synthetic-data pipeline actually makes.

3.5 Decision 1 — deduplication is a corpus instrument, not an item gate

Deduplication reads the closest pairs, which is where whatever signal exists is concentrated, and at corpus scale it works: duplicate-pair detection on the code corpus runs at **AUC 0.779** against the ground-truth label *identical on the whole battery*.

Per item it does not. Among the pairs a near-duplicate filter would reject, the fraction that are behaviourally *far apart* (distance > 0.5):

filter radius	pairs rejected	of those, still behaviourally far	mean behavioural distance
5th pctile	1,026	**45.4%**	0.419
10th	2,051	62.2%	0.549
20th	4,101	76.8%	0.688
30th	6,151	83.0%	0.751
40th	8,201	86.3%	0.784
pool-wide			*0.853*

At the tightest radius, nearly half of what the filter discards behaves unlike the item it was discarded for resembling; the best radius available reaches F1 0.449 (precision 0.452, recall 0.446). A second decoder — 202 strings scored by the `ipaddress` arcs they execute — gives 41.6% at the same quantile against a pool-wide mean of 0.672.

This is a direct measurement of the decision, so the §3.3 control does not touch it. Threshold tuning does not fix it either, because the error is in the strength of the evidence rather than the location of the boundary. **A text-space duplicate filter is defensible as a corpus-level instrument — a duplicate rate, a contamination screen — and not as a per-item accept/reject.**

3.6 Decision 2 — diversity selection has no stable sign, and both signs lose to the oracle

Farthest-point traversal draws its picks from the top decile. What that costs turns out to depend on the decoder, and not in a way anything visible in the text predicts. On the `ipaddress` decoder, greedy max-min is worse than random at every budget in every representation, by 54 to 82 arcs. On the code corpus it is worse than random at $k = 10$ and better from $k = 20$ on:

k	random	max-min	best near-duplicate filter	**artifact-space oracle**
10	815	743	849	**1,336**
20	1,119	1,296	1,267	**2,047**
30	1,355	1,509	1,517	**2,365**
50	1,549	1,864	1,809	**2,414**
100	1,891	2,358	2,113	**2,414**

Behaviour cells covered, mean of 20 seeds; the oracle is greedy marginal coverage computed in the artifact's own space.

Two things follow. **Whether text-space diversity selection helps is a property of the decoder and is not predictable from the embedding** — two decoders, the same selector, opposite signs. And **the gap that matters is against the oracle, not against random**: the best text-space selector at k = 10 buys 849 cells where the oracle buys 1,336, the oracle reaches at k = 30 what text-space selection needs k = 100 to reach, and it saturates the pool at k = 50 where no text-space arm saturates at all. Text-space selection leaves between a third and a half of the reachable behaviour unbought at small budgets whether or not it beats random.

The oracle is normally waved away as the thing one cannot afford. §8 measures what it costs.

3.7 Decision 3 — the redundancy that no text-space instrument reaches

The corpus itself makes the case for measuring the artifact more compactly than any correlation does. Naive repeated prompting produced 553 programs. Removing byte-identical sources — the deduplication step every pipeline performs — leaves **203**. Those 203 distinct programs exhibit **39 distinct behaviours**.

So text-level deduplication, performed perfectly, removes 63% of the corpus and leaves it **5.2× redundant in the space that matters**. The residue is not subtle paraphrase. One pair of programs in it implements the longest-increasing- subsequence length by patience sorting with a binary search, and the other by the quadratic dynamic program; they share almost no tokens, and they agree on all 140 battery inputs. No embedding-space threshold reaches that pair without also discarding most of the corpus, and running both programs takes microseconds.

4. The lever: the same channel, used to instruct

Everything in §3 uses the control modality as a *ruler* — a space in which to measure how far apart two items are — and every use of it in that role failed to clear a control. This section uses the same channel for the other thing it can do: carry an instruction. The permutation nulls of §3.3 say the information is present. The question is whether it can be put in rather than read out.

4.1 Stating the target, and checking by execution

A **target** is one behavioural cell: a required output for a specified input, such as *`f([1, 2, 3])` must return 7*. It is a fact about the artifact, and it can be written into the text. Whether the artifact then satisfies it is settled by running the program, not by asking the model.

Four arms, matched at one generator call each, 48 targets × 3 seeds:

arm	what the prompt says	compliance	95% CI
`blind`	the fixed prompt; the target is never mentioned	18.1%	[12.5, 24.3]
`decoy`	a *different* target, stated in identical form	**6.9%**	[2.8, 11.1]
`instruct`	the target	**62.5%**	[54.2, 70.1]
`retry`	`instruct`, then re-prompt showing the wrong value	61.8%	[54.2, 69.4]

The decoy arm is the one that makes this a result rather than an observation. Adding any concrete requirement to a prompt changes what a generator produces, so `instruct` beating `blind` would be consistent with the channel carrying nothing but specificity. It is drawn from the same input index wherever possible, so the decoy and instruct prompts differ in exactly one number.

The decoy scores below blind prompting — 6.9% against 18.1%. Stating the wrong target does not merely fail to help; it actively moves the generator away from the right answer, well below the rate it reaches with no instruction at all. The channel transmits *which* target, not that a target exists.

Re-prompting buys nothing: 61.8% on 149 calls against 62.5% on 144. Showing a model its own wrong answer and asking again does not recover the misses, which suggests the failures are targets the generator cannot reach rather than ones it carelessly missed.

4.2 How far into the tail the lever reaches

A single compliance rate would hide the structure that matters. Each target's **rarity** is the fraction of the generator's own natural corpus — 203 distinct programs written under the fixed prompt — that already satisfies it. Rarity 0.35 is a target one program in three hits by accident; rarity 0.005 is one that a single program in the corpus happens to hit.

rarity of the target	`blind`	`decoy`	`instruct`	lift over decoy
[0.30, 1.01)	58.3%	16.7%	**100.0%**	+0.833
[0.10, 0.30)	5.6%	5.6%	**63.9%**	+0.583
[0.02, 0.10)	5.6%	2.8%	**55.6%**	+0.528
[0.001, 0.02)	2.8%	2.8%	**30.6%**	+0.278

The lever works across the whole range and its strength is a function of where the target sits in the distribution being steered into. In the head it is certain. Four bands out into the tail it still triples the decoy rate ten times over, and still misses two targets in three.

This is the engineering quantity a practitioner needs, and it is not the quantity a diversity metric reports. It says: you can put a specific artifact- space requirement into the control modality and expect it honoured with a probability that depends on how unusual the requirement is for this generator. Both halves of that sentence are actionable — the first licenses targeted generation, the second says how much budget to allocate per target and where a generator will need help.

Three caveats belong with the number. It is measured on one decoder. Compliance is checked on the commanded input only, so a program that satisfies the target by special-casing that input would count as a hit, though the prompt forbids it. And the rarest band the corpus can supply is 1/203; targets the generator has *never* produced are the most interesting point on this curve and are not yet measured.

5. Measure with the decoder: it is cheaper than the channel

The argument for computing distances on text rather than on the artifact is always the same, and it is never stated as an argument because it is assumed: decoding is expensive, embedding is cheap, so one embeds. For a rendered image or a two-minute instrumental that is true. For the decoders in this section it is false, and by a wide margin.

Measured on one laptop, per item:

operation	ms per item	relative
decode: string → `ipaddress` arcs, plain execution	**0.11**	1×
decode: same, with `sys.settrace` instrumentation	**0.98**	9×
decode: Python program → 140-input battery, **each in a fresh OS process**	**29.1**	265×
embed: `nomic-embed-text`, local Ollama, batched	107.9	981×
embed: `nomic-embed-text`, local Ollama, one at a time	1,488.2	13,529×
embed: character TF-IDF, local, batched	18.3	166×

The code row is deliberately the least favourable decode measurement we can honestly make: every generated program is run in a *separate operating-system process* for isolation, so Python interpreter start-up dominates and the actual execution is a rounding error. Even so, decoding is **3.7× cheaper**

than a batched local embedding and **51× cheaper than embedding items one at a time**, which is what a streaming pipeline does. Where the decoder does not need process isolation the margin is three orders of magnitude.

Two honest qualifications. Instrumentation, not the program, is the dominant decode cost — tracing costs 9× plain execution — so a decoder whose instrumentation is heavy needs its own measurement rather than an inherited ratio. And the embedding figures are for a *local* model with no network hop and no per-token charge, which is the best case for the text-space route; a hosted embedding endpoint is slower and has a price.

The conclusion is not rhetorical. **For a deterministic decoder that runs in milliseconds, the artifact-space oracle is not the expensive option a practitioner forgoes. It is the cheap option they did not think to run**, and it recovers between 1.02× and 1.64× the coverage of the best text-space selector at matched budget (§3.6), reaching at $k = 30$ what text-space selection needs $k = 100$ to reach.

Where decoding genuinely is expensive — rendered images, audio, anything behind a paid endpoint — the fallback is not a text-space selector but a *partial decode budget*: decode a sample, and spend the rest of the budget where the sample says the behaviour is. That regime is where §5's cross-modal steering belongs, and it is the boundary between the two halves of this paper.

6. `decodergap`: the probe

The results above do not compose into a rule a practitioner can apply blind, because the one decision whose sign varies — far-field selection — varies *by decoder*, and nothing visible in the text says which case you are in. What they compose into is a measurement, and the measurement is cheap enough to run before any pipeline decision is committed.

`decodergap` takes three things: a corpus, an embedding function, and a decode function returning anything with a distance on it. It returns the near-field profile, the two deduplication verdicts, and the selector comparison against the artifact-space oracle.

```
from decodergap import probe

report = probe(
    texts     = corpus,                     # what you generated
    embed     = my_embedder,                # what you were going to select in
    decode    = lambda s: run(s, battery), # what the item actually becomes
    distance  = jaccard,                    # a distance on the decoded artifact
    coverage  = distinct_cells,             # what a subset is meant to buy
)
report.verdicts["deduplication_aggregate"]   # SUPPORTED | UNSUPPORTED
report.verdicts["deduplication_per_item"]    # SUPPORTED | NOT SUPPORTED
report.verdicts["max_min_selection"]         # SAFE | UNSAFE, and vs the oracle
```

Three design decisions carry the tool, and each is a finding from above rather than a preference.

It never reports a correlation without its decoder-free control. The profile of §3.2 looks like a finding and is reproduced twice as strongly by two encoders that never saw a program run (§3.3). Any proxy-qualification number the tool emits is accompanied by the encoder-to-encoder baseline it has to beat, and by the permutation null that shows whether any association exists at all.

It issues two deduplication verdicts, not one. The aggregate verdict answers *can this radius rank duplicate pairs at all*; the per-item verdict answers *of the pairs a filter at this radius would reject, how many are behaviourally far apart*. On both decoders measured here the first passes and the second fails, at 45.4% and 41.6%. A tool that emitted a single ”dedup: safe” would be licensing the decision its own data refuses.

It scores every selector in the artifact's space and against the oracle, so a text-space selector cannot win by agreeing with the representation it selected in, and beating random is not mistaken for doing well.

It also refuses to recommend a filter radius. We swept the threshold over the 5th to 40th percentiles of each corpus's own distance distribution at four budgets: the best cell on the `ipaddress` decoder is +20.7 arcs over random at t = 10th percentile and $k = 10$, and by $k = 50$ every threshold loses. Any single-budget experiment would have produced a shippable-looking constant. The sweep is reported in full, including the cells that lose, and no constant is offered.

7. When the decoder is expensive

Everything above rests on decoding being cheap. Where the decoder is a diffusion model or a two-minute audio generation, the oracle of §5 is genuinely out of reach and the control modality is all a pipeline has. This section reports what the same questions look like there, on the corpora of the companion paper. The answers are worse, and they are worse in the direction §3 predicts.

7.1 Text diversity is nearly blind to image diversity

We rendered the first 199 DALL·E instructions from the naive corpus and embedded them with CLIP. Pairwise cosine similarity in text-embedding space correlates with pairwise cosine similarity in image-embedding space at **Pearson *r* between 0.167 and 0.421** across the seven rendered arms, and at 0.285 within-arm centred. The gap varies by a factor of 2.5 depending on which corpus is measured: lowest on `self_instruct` (0.167) and on the naive arm (0.170), highest on the arm this paper's own method produced (0.421). Pooled, the shared variance is 14.5%; within-arm it is 8.2%. Knowing that two instructions are semantically far apart therefore tells you rather little about whether the two pictures look different, and every text-side diversity method in this literature — ours included — is optimizing a proxy that leaves most of what the reader receives unexplained. That the correlation is strongest on our own corpus is worth stating plainly: the proxy is least misleading exactly where the instructions were built to differ structurally rather than only lexically.

The qualitative version is more damning than the correlation. Independently generated instructions, from a corpus with a 0.000 exact-duplicate rate and healthy lexical diversity, render to near-interchangeable pictures: the same magenta-and-cyan palette, a classical marble bust, neon signage, halftone collage, a receding grid. A vision judge shown samples rates the set's distinctness 6–7 out of 10 and names the attractors precisely — *"muted beige, cream, brown, ochre and black foundations accented by saturated cyan/teal, turquoise, pink"*, *"appropriation of canonical or religious imagery, especially Mona Lisa-like female portraits"*, *"frontal, museum-like presentation with centered, symmetrical compositions"*. This is a second mode collapse, downstream of ours, contributed by the image model and by the fact that much of what varies in the text lands in the same visual place.

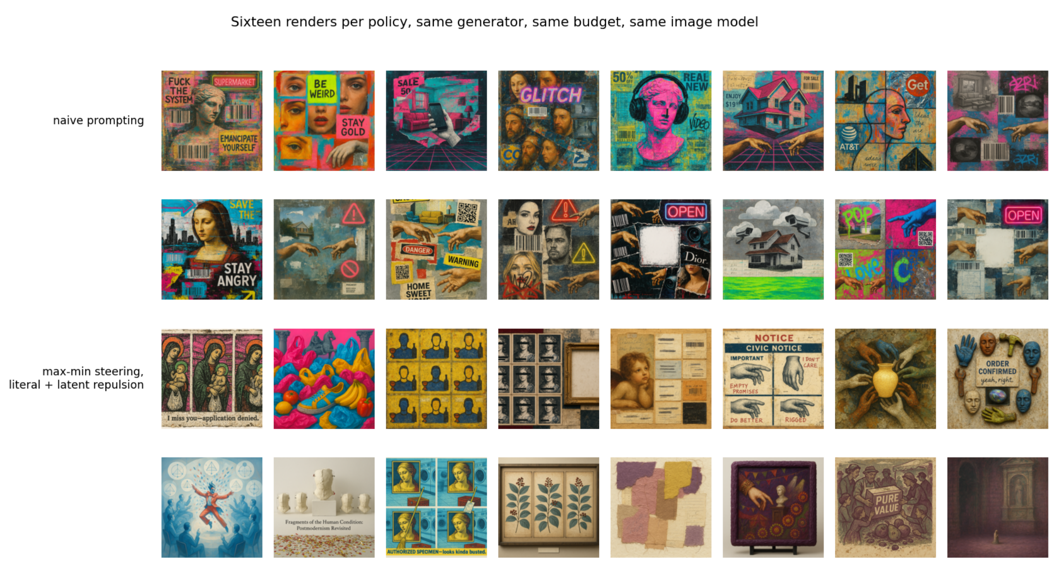

Figure 1. Sixteen renders per policy, same generator, same budget, same image model. Rows 1–2, naive prompting: a corpus with a 0.000 exact-duplicate rate and healthy lexical diversity that still returns one visual mode — magenta and cyan collage, barcodes and QR codes, Renaissance portraits

and classical busts, Michelangelo hands, warning triangles and neon OPEN signs recurring across nearly every panel. Rows 3–4, max-min steering with literal and latent repulsion on both the text and vision sides: medium, palette, register and composition all move, across a medieval triptych, a botanical cabinet, a photographed sculpture installation, a torn-paper abstract and a civic notice.

7.2 Audio: whether a prompt can be steered is a property of the embedder

The more portable finding is that whether a prompt can be steered before rendering is a property of the *embedder* rather than of music. Measuring the Spearman correlation between prompt-pair similarity through the text tower and track-pair similarity through the audio tower:

embedder	alignment (naive / conditioned)	note
MuQ-MuLan	**0.68 / 0.47**	a usable steering gradient
CLAP-fused	0.50 / 0.21	weak; ~0.05 on the steered corpus
CLAP-music	0.18 / 0.06	worst; median within-corpus NN distance 0.004 — it hears one track

Cross-embedder agreement on pairwise track similarity spans 0.22–0.84, per-arm diversity verdicts flip between embedders, and each embedder nominates a different most-redundant pair. Two prescriptions follow: steer music with MuQ-MuLan rather than CLAP, and never publish an audio-diversity number without naming its embedder. Structural contracts have a length ceiling as well — compliance with a requested section sequence is exact at prompt lengths near 560 characters, 0.11 at a mean of 658, and zero by ~1,600 — so a structural contract for music must be short or it is noise.

7.3 The ranking does not depend on the viewpoint

A max-min score is a distance in a chosen embedding, so a natural worry is that it names a property of the embedder rather than of the corpus. Six exam-item corpora — this method's bank against naive sampling, persona prompting, self-instruct, evol-instruct and high-temperature sampling, 200 items each — were scored under four independent representations: CLIP text, TF-IDF with no learned semantics at all, the twenty-two-feature prosodic vector, and a 150-dimensional function-word profile of the kind used for authorship attribution. Each score is the fifth-percentile nearest-neighbour distance divided by the mean pairwise distance of the same cloud, which is invariant to the global rescaling an embedding choice is free to apply, and each representation is admitted only if its between-corpus spread exceeds its own within-corpus sampling noise — all four clear that bar by factors of 9.7 to 24.7.

The four views agree, at a mean Kendall tau of +0.83 across their six pairings, and **RAC ranks first under every one of them**, including the three it never optimizes. It also holds the best worst-case score, so it wins under a distributionally-robust reading of diversity as well as under any single view. The agreement is not an artifact of the degenerate baselines: restricted to the three corpora whose fifth-percentile distance is non-zero in every representation, mean tau is +0.67 and RAC is still first under all four. Naive sampling and high-temperature sampling score exactly zero on prosody and on stylometry — at least one item in twenty has an exact neighbour in those channels.

7.4 A class of repetition the embedding misses

One comparison in the artifact channel runs against the method, and it belongs here. Embedding metrics miss an entire class of visual repetition — tiled grids of one cell, a shared palette, the same composition recoloured — so we audit it with deliberately dumb non-semantic signatures: a 16×16 luminance layout map, an autocorrelation tiling score, a hue histogram. Scored at matched n = 60 and matched render tier, **the published baselines are better than our arms on both structural measures**: layout twins above 0.5 cosine average 0.223 across the five baselines against 0.473 across our eleven, and literal tilings 0.117 against 0.406, with no baseline worse than our best arm on layout. A corpus told to differ along seven or eleven named dimensions apparently converges on a compositional template that plain repeated prompting does not. Adding literal-space structural bans to the prompt — grids banned when recent renders tile, dominant hue pairs named and banned, layout-change demands when layouts collide, plus a learned text→bad-structure bridge — halves palette twins (0.78 → 0.35 in the coverage arm) and improves the coverage objective 21% (0.206 → 0.247), but does not close it. Optical spread and structural repetition are both measured on pixels and point in opposite directions: our corpora occupy a wider region of that space while repeating their compositional scaffolding more often within it.

7.5 A defect no diversity measure can see

The audits of §6 ask whether the axes reach the artifact. This one asks whether the resulting bank is *usable*. For a test bank, key position is a hard requirement: correct answers must be distributed across option positions, because an examinee who notices an imbalance can exploit it without reading anything. Asking a fixed solver each item and recording which position it chose:

bank	A	B	C	D	χ^2 vs uniform (3 df)
RAC	**0.900**	0.050	0.050	0.000	**90.4**
naive	0.225	0.550	0.225	0.000	24.6
RAC, after balancing	0.200	0.175	0.300	0.325	**2.6**

Ninety percent of the RAC bank's keys sit in position A, against a critical value of 7.81 at $p = .05$. An examinee who answers A to everything scores 90% without reading a stem, and neither bank ever places a key in D. This is a property of the solver only if the solver is not reading the items, so we separated the two by permutation: ask each item twice, once as written and once with its options reordered, and record which option *text* is chosen. The answer follows the content in 95% of RAC items and 100% of naive items, and stays on the same letter in only 17.5% and 30%. The solver reads; the imbalance is in the bank.

Why the method cannot see this. Key position is not a semantic property, so it is invisible to every measure in this paper — embedding diversity, n-gram diversity, Vendi, coverage and the enemy-item radius are all exactly as favourable on a bank whose key is always A as on a balanced one. It is invisible to the axis set for the same reason: all seven axes condition on item content, and even *distractor logic* governs what the distractors are like rather than where the key sits among them. And it is structural rather than accidental — a generator asked for a stem and four options writes the answer it has in mind first and builds distractors around it, so A is where the key lands by default.

The repair needs no answer key. Permuting each item's options uniformly at random moves the key with its own text from whatever position it held, so the bank becomes uniform by construction; options whose text refers to a position (”none of the above”, ”both A and B”) are detected and left as written, which affects 37 of 1,999 items. Applied to the RAC bank this takes χ^2 from 90.4 to 2.6 — indistinguishable from uniform — while content-tracking is unchanged at 95%.

The general form is the sharpest limitation in this paper. **A diversity objective defined over an embedding is blind to any property of the artifact the embedding does not encode, including properties that decide whether the artifact can be used at all.** Difficulty is the same story in the same domain: it is the parameter an item bank exists to span, it is latent rather than semantic, and nothing in the objective can see it. Measuring diversity well is not the same as measuring fitness for purpose, and the gap between them is not visible from inside the objective.

7.6 Steering through the proxy where it is all you have

7.6.1 Images: closing the loop where the product lives The vision-steered arm closes the loop where the product lives. It renders a bounded sample of accepted instructions, embeds them with CLIP, and feeds two things back into the text-side loop: a least-squares map from the crowded *image* directions into instruction-embedding space, so the text-side orthogonality term can push away from visual redundancy it cannot itself perceive; and mined *visual* attractors from the vision judge, appended to the same ledger as the textual ones. It is the paper's mechanism applied one level down — the ledger already repels against what the model keeps saying, and now also against what it keeps showing — and it is the best arm in the companion paper's competitive comparison (its §7.2).

The margin is measurable in the space the loop optimizes and visible on the page. At matched n = 200 in the DALL·E domain, the text-only RAC arm scores centered Vendi 77.67 and median nearest-neighbour distance 0.171; the vision-steered arm scores 91.97 and 0.192, an 18% gain in centered Vendi, and leads every column of the seven-arm comparison — distinct-2 0.699 against 0.660, 4-gram self-repetition 0.040 against 0.068, n-gram Vendi 170.9 against 167.0. The contact sheet of §4.2 shows the same thing without a number: rows 3–4, steered on both the text and the vision side, move medium, palette, register and composition where rows 1–2 return one visual mode.

7.6.2 Audio: cross-modal steering through an audio embedder At matched n, matched prompt length and the same generator, the conditioned music arm reaches centered Vendi 43.03 against naive's 22.99 (1.87×), halves 4-gram self-repetition (0.140 against 0.291), raises distinct-2 (0.545 against 0.348) and holds roughly three times the room between nearest neighbours (0.079 against 0.027), on 100 prompts per arm with no exact duplicates in either — so the effect is semantic. On 100 rendered Lyria instrumentals generated by cross-modal steering with zero rejection, 100% of tracks verify as instrumental in CLAP space and mean-centered CLAP Vendi is 17.43 against 11.5 for naive prompts and 14.1 for axis-conditioned prompts under the same embedder.

The rendered arm was steered through CLAP and is measured in CLAP. §7.2's alignment table says the same loop through MuQ-MuLan would have a steering gradient nearly four times as strong on the naive arm (0.68 against 0.18), which is the experiment we would run next; the prescription — steer music with MuQ-MuLan rather than CLAP — is §7.2's, and the loop is the image one of §7.6.1 with the audio tower in place of CLIP.

7.6.3 Best-of-K at the render: select on the channel the objective cannot see The companion paper measures best-of-K on the written candidate (its §8.4): selecting on the gap alone raises the minimum nearest-neighbour distance by 20% in three of three seeds, at a cost of a third of a craft point. The same question at the *render* has a sharper answer, because the two channels have very different effective dimensions. Best-of-K on the written candidate happens before the handoff and §6 locates the loss after it, so we also ran $K = 3$ renders behind each of 60 fixed instructions and kept the render maximizing min distance to the accepted set — once in CLIP, once in optical statistics, over the same candidates. Selecting on optics raises pixel min NN from 1.039 to 1.292 (1.24×) and the fifth-percentile NN by 0.336 (paired subsample, 95% CI [+0.128, +0.488]), **and costs nothing in the channel the method optimizes**: CLIP min NN moves by −0.001 with the interval [−0.029, +0.033] tight around zero. Selecting on CLIP raises CLIP min NN by 0.040 ([+0.025, +0.071]) and moves optics not at all detectably. Both gains lie on the $K^{1/m}$ curve — 1.24× at $K = 3$ implies $m \approx 4.4$, and 1.04× implies $m \approx 17$ — so render-side selection is oversampling rather than a new mechanism. What the dimensions add is an allocation rule: oversampling buys far more in a low-dimensional channel than in a high-dimensional one, and **if a pipeline is going to pay for extra renders, it should select them on the channel the objective cannot see.**

7.7 Conditioning across the seam

The companion paper audits whether a commanded axis level shows up in the artifact (its §8.1), by manipulation rather than correlation: hold a base spec fixed, set one axis to each of its levels in turn, generate, and read the displacement, centring within base so that only variation produced by *setting* the axis contributes, with a null that shuffles levels within each base. Two instruments read each artifact — *identifiability*, whether CLIP(ℓ) is the nearest of the axis's level descriptions to an image generated under level ℓ, against a chance rate of 1/|levels|; and *separation*, the η^2 of artifact embeddings grouped by commanded level under a permutation null — and a third channel that lives outside the embedding the method optimizes: pixel statistics for images, prosody for poems. On poems every axis is realized on at least one channel. On images three of seven axes identify their level at or below chance, two are undetectable on both channels, and the axis the attribution scoring ranked *first* is the least realized in the corpus. The image domain is where the proxy question becomes a conditioning question, because the image pipeline has a seam the poem pipeline does not.

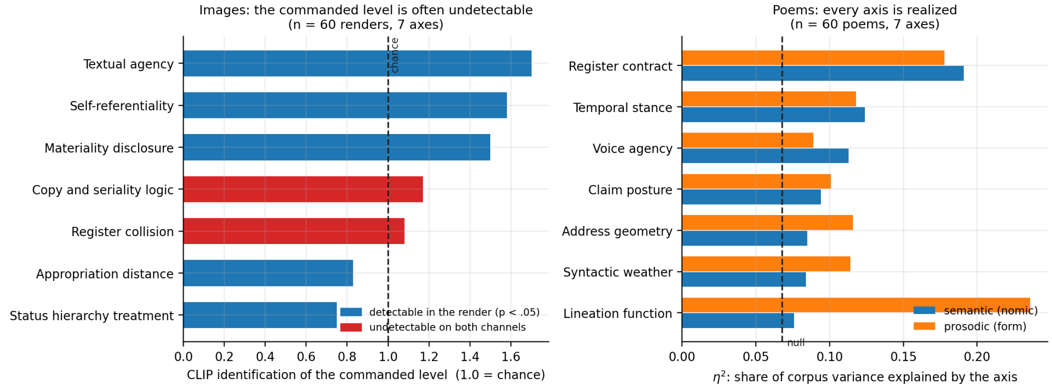

Figure 2. Axis realization measured on the artifact. Left: on 60 max-min renders, three of seven axes identify their commanded level at or below chance, and two are undetectable on both the CLIP and pixel channels. Right: on 60 poems, every axis is realized — on the semantic channel, the prosodic channel, or both. Dashed lines mark chance and the permutation null.

Images, by intervention: the loss is at the render, not in the writing. The image pipeline has two stages — the model that proposed the axes writes an instruction, and `gpt-image-1-mini` renders it — so the same manipulation can be read on both sides of the handoff. Over all eleven axes at three bases and five levels, embedding the written instruction with CLIP, the render with CLIP, and the render's optics separately: nine of eleven axes are realized in the **written instruction** at $p < 0.05$ (mean $\rho = 0.148$), against four of eleven in the **render** (mean $\rho = 0.096$). Ten of the eleven attenuate, a mean loss of 35% of the text-side effect (Wilcoxon signed-rank $p = .005$; paired t, $p = .003$, both two-sided). **Conditioning reaches the instruction and loses a third of its grip at the render.** The image domain's weak realization is therefore not a failure of the axis calculus to produce usable conditioning — the conditioning is there, in the prompt, and measurable — but of that conditioning to survive a generator that never saw the axis set.

Part of the apparent loss is the instrument rather than the renderer, and this is the §4.1 lesson arriving from the other direction. The four perceptual axes attenuate most in CLIP and are precisely the axes CLIP is least equipped to register: measured on pixels instead, *Lighting direction* reads 0.229 against 0.092 in CLIP, *Shot scale* 0.252 against 0.105, and *Motion and temporal blur* 0.203 against 0.069. An axis commanding optics moves the optics and barely moves a semantic embedding.

The seam explains the split. The calculus is identical across the two domains, the axes are comparably abstract, and the budgets match. What differs is where the conditioning has to travel. A poem is written by the same model that proposed the axes, so the conditioning never leaves the system that understands it. An image axis is written by that model into a prompt and handed to a different model, which never saw the axis set, shares no latent space with it, and is under no obligation to honour a distinction like *copies outrank an absent original*. §4.2 measured this seam from the outside as a proxy gap; the intervention measures it from the inside. **Conditioning strength degrades at the boundary where the generator changes hands**, so the modality-agnostic claim holds for the calculus — which needs only an embedding — and not for the conditioning, which needs a generator that reads the same language the axes are written in.

8. Practice

The prescriptions are short, and each is attached to the measurement that motivates it. The first four are the ones we would keep if we could keep four.

1. **Never report a proxy-validation correlation without a decoder-free control.** Correlate your embedding against a second, unrelated representation of the same corpus, and against a permuted version of your artifact measurement. Whatever agreement survives above the first is what the artifact contributed; the second tells you whether any association exists. On our corpus the encoder-to-encoder baseline was **twice** the decoder signal (§3.3), which retires the finding we had.

2. **Qualify a proxy by running the decision, not by correlating the distances.** A correlation summarizes a geometry and inherits that geometry's confounds. A decision is scored in the artifact's space and either works or does not: what fraction of the pairs your filter drops are genuinely duplicates, how much artifact coverage your selector buys against the oracle.
3. **Split every deduplication claim into a corpus claim and an item claim.** They have different evidence and on both decoders here they get different verdicts: AUC 0.779 for ranking duplicate pairs, and 45.4% of rejected pairs behaviourally far apart at the tightest radius (§3.5). Contamination screening is defensible; a per-item gate is not.
4. **If your decoder is deterministic and fast, do not select in text space at all.** Decoding cost 29.1 ms per item against 107.9 ms to embed one, in the least favourable measurement we could construct (§5), and the artifact-space oracle reached at $k = 30$ what text-space selection needed $k = 100$ to reach (§3.6). The oracle is the cheap option, not the expensive one.
5. **Name the embedder, and publish the pairwise-similarity distribution it operates on.** Per-arm audio verdicts flip between embedders and each nominates a different most-redundant pair (§7.2); mean pairwise cosine was 0.883 in one text corpus and 0.444 in another under the same embedder. A diversity number without its embedder and its similarity distribution is not a number.
6. **Count exact duplicates before computing anything, and report the literal and the latent separately.** They move in opposite directions in the same corpus (§7.3), and a 74% duplicate rate makes every distance statistic a statistic about duplication.
7. **Measure at the level of the artifact you ship.** Text-embedding similarity explains 14.5% of the variance in whether rendered images look alike, pooled across seven arms, and 8.2% within-arm (§7.1). If the artifact is rendered, the objective belongs in the space the artifact occupies.
8. **Audit in at least one channel the objective never optimizes**, with a distance statistic rather than a spectral one (§7.1). It is the only measurement that cannot be a restatement of the selection rule, and it is where the strongest positive result and the largest undetected defect in this line of work were both found.
9. **When a render is in the loop, steer through it.** Render a bounded sample, embed it in the artifact's space, and bridge its crowded directions back into the text space the loop optimizes (§7.6.1); steer audio through an embedder whose text and audio towers agree (§7.6.2).
10. **Spend extra renders on the low-dimensional channel.** Best-of-K buys $K^{1/m}$; selecting renders on nine pixel statistics buys 1.24× at $K = 3$ where selecting on CLIP buys 1.04× (§7.6.3).
11. **Measure axis realization at the artifact, by intervention, wherever the generator changes hands** (§7.7). Where the seam exists, a third of the conditioning does not cross it.
12. **Balance by construction what the embedding cannot see.** Key position, difficulty, compositional template: none is encoded, none is optimized, and each decides usability. A uniform permutation of options needs no answer key and takes χ^2 from 90.4 to 2.6 (§7.5); a structural ban block halves palette twins (§7.4).
13. **Read the output.** A corpus can be diverse along every dimension you thought to measure and uniform along the one you did not.

9. Limitations

- **One text embedder.** All text geometry is `nomic-embed-text` with cosine distance. §7.3 shows the *ranking* of methods survives four representations; whether the *magnitudes* transfer is untested, and the companion paper's coverage and packing numbers inherit that.
- **A joint embedder varies both towers at once.** §7.2 changes one knob — the model supplying both the text tower and the audio tower — so it shows that the alignment moves and cannot say which side moves it. Crossing text-side and artifact-side encoders independently on fixed bytes is the decomposition this paper does not contain.
- **The renders carry no logged seed.** The image endpoint exposes no seed parameter, so no render in this paper can be regenerated, and render stochasticity cannot be separated from the quantity being measured. Every artifact-space number here — the proxy correlations of §7.1, the realization of §7.7, the render selection of §7.6.3 — is a statement about an unlogged seed

distribution, and a repeat-draw floor at a logged seed is the missing measurement.

- **Several image results are single-seed.** The image arms are one seed each at $n = 60$; the vision-steered arm's margin in §7.6.1 is one run at $n = 200$; the audio steering arm is one run of 100 tracks.
- **Judge bias is unmeasured.** Vision and language-model judges name attractors, classify blueprint areas and read device forms; none is calibrated against human raters, and a judge that shares the generator's blind spots would miss exactly what the generator misses.
- **The structural signatures are deliberately dumb.** A 16×16 luminance map, a tiling score and a hue histogram find tilings and palette twins and nothing subtler; they are a floor on what the embedding misses, not a measure of it.
- **Steering closes part of the gap and not all of it.** Literal-space bans halve palette twins and improve the coverage objective by 21% and do not close the structural deficit; the vision-steered arm is the best arm in the image domain and still renders through a model that never saw the axes.

10. What this cost us to learn

The confound of §3 is not an exotic failure mode. It is the default, and we fell into it repeatedly while writing the paper that describes it. Sorting our own results by whether they survived scrutiny makes the pattern hard to miss.

Survived — every one a direct measurement of a decision, scored in the artifact's space:

- the lever result and its decoy control (§4)
- the partial-decode budget curve, and the oracle intervals (§5)
- the cost comparison (§5)
- reject purity of a near-duplicate filter, 45.4% and 41.6% on two decoders (§3.5)
- max-min's standing against random, and the fact that its sign flips between decoders (§3.6)
- the decoder-free control itself, implemented independently in two sessions on different decoders within the same hour, agreeing

Died — every one an analysis of the shape of a correlation:

- the near-field law as a property of decoders, killed by the control in §3.3
- its apparent invariance across four encoders, which was the same artifact seen four times
- a within-decoder version of it, which survived until a permutation null put it at $p = 0.17$ — 17% of arbitrary splits of the same corpus produce a gap as large
- a refinement claiming dispersion rather than mean smoothness was the operative variable, which ordered the data backwards
- an early cross-representation result that a separability gate later showed could not have been measured at all, because three of five views could not rank the corpora above their own sampling noise

Five failures, one survivor class. The failures were not bad luck: correlation shape is precisely the evidence type this paper argues is confounded, and we kept reaching for it because it is cheap, because it produces a table quickly, and because a decile profile with a monotone trend and $p = 3\times10^{-103}$ in its first cell looks like a finding.

One failure is worth stating in detail because the rule that catches it is general. The SQL decoder appeared to be the one case that cleared its baselines. It cleared *two* of them — the cross-family encoder pairs — and we discounted the third, the lexical-versus-lexical pair, as too easy a baseline to be meaningful. That was a justification constructed after seeing which baselines were inconvenient. Under the rule that a decoder must clear the whole baseline *distribution*, the margin is −0.195, 95% CI [−0.337, +0.013].

> **Choose your control before you see it, and clear the whole distribution of controls rather than a member you have selected.**

We would not have accepted the discounted baseline from anyone else, which is the usual sign.

11. Conclusion

You can address the generator in one modality and you need the output in another. The channel between them is real: permuting the artifacts destroys the association in every domain we measure, so text and artifact are genuinely coupled, and a corpus built by writing text is not writing into the void.

What the channel will not do is serve as a ruler. Distances measured in it do not stand in for distances in the artifact, the standard check that would catch this is confounded by a shape two arbitrary encoders produce between them, and the decisions built on those distances come apart under direct measurement — a duplicate filter that is sound for a corpus statistic and unsound per item, a diversity selector whose sign depends on a decoder nothing in the text reveals.

What it will do is serve as a lever. A specific artifact-space requirement, written into the text, is honoured 62.5% of the time against 6.9% for a decoy that states a different requirement in the same words — and the wrong requirement leaves the generator worse off than no requirement at all, which is how we know the channel is carrying the content rather than the form. How often it is honoured depends on where the target sits in the generator's own distribution: certain in the head, one time in three four bands into the tail.

The practical shape of that split is short. Steer through the channel, because that is the use it supports. Measure with the decoder, because for anything deterministic the decoder is cheaper than the embeddings it replaces and buys 1.54× the coverage at small budgets. And when you must qualify a proxy by correlation because the decoder is a diffusion model and you cannot afford it, run the decoder-free control first, on a baseline you chose before you saw it.

We did not run that control until the fifth attempt, and it retired four of our own results. That is the strongest argument we can make for it.